

Crowdsourced Delivery Problem With Couriers' Preferences: A Data-Driven Method With Logic-Based Benders Decomposition

Xuting Sun¹, Yue Hu, and Jiage Huo²

Abstract—With the rapid development of food delivery platforms, crowdsourced delivery mode has become popular due to its cost-effectiveness and flexibility. The crowdsourced couriers can choose to accept or reject orders dispatched from the platform. Such behavior not only affects the income of the couriers and the customer service quality, but also brings about negative impacts on the platform's operational efficiency and its reputation. It is essential to analyze the preferences of the couriers so that the platform can develop a “courier-friendly” and high efficient delivery dispatching strategy. Therefore, our study first uses a real-world food delivery dataset to identify the factors that influence crowdsourced couriers' willingness to accept orders. In particular, we apply an extreme gradient boosting tree (XGBoost) to forecast the likelihood of the crowdsourced couriers accepting the delivery tasks. We further integrate the likelihood of order acceptance into a mixed-integer programming model to optimize the dispatching decisions for the platform. A multi-criteria objective function is proposed aimed at making a trade-off between the labor cost, service quality as well as couriers' welfare. To solve the problem, a logic-based Benders decomposition algorithm is proposed, and an efficient analytical cut is developed to accelerate the algorithm. The numerical results verify that the proposed solution approach has significant advantages in terms of computational efficiency and solution quality compared to the off-the-shelf solver. Moreover, it is identified that, for on-demand delivery platforms, integrating courier preference information into the courier dispatch system is a vital strategy for achieving more efficient and cost-effective operations, as well as an important pathway to achieve social sustainable operations.

Index Terms—On-demand food delivery, crowdsourcing, willingness to accept orders, machine learning, logic-based Benders decomposition.

I. INTRODUCTION

WITH the rapid development of e-commerce and local life services, on-demand food delivery, which is a typical online-to-offline (O2O) service [1], has become a critical component of urban logistics. With the O2O service, customers can choose products and place orders online, and receive the ordered products offline [2]. The O2O platforms

serve as the foundation for on-demand food delivery, offering fast point-to-point delivery services from stores to customers to satisfy the demands of online consumers [3]. By June 2024, the global user base of food delivery services has exceeded 3 billion,¹ accounting for approximately 60% of all netizens worldwide. In the international market, multiple food delivery platforms have rapidly expanded. For example, Uber Eats, one of the most widely used food delivery platforms, generated almost \$12.2 billion in revenue in 2023.² DoorDash, the market leader in the US, has achieved 60% of the US food delivery market share by 2023.³ In China, Meituan and Ele.me hold more than 90% of the domestic food delivery market.⁴

With their large user base and efficient distribution network, these food delivery platforms are pushing the industry forward. Major food delivery platforms are continuously enhancing their delivery time guarantees, with standard service windows typically ranging from 30 minutes to an hour. Some platforms have also introduced “Flash Delivery” services. In addition, there are also compensation plans for delivery delays on various platforms to guarantee the customer experience. The compensation amount is decided by the order value and delay duration, typically ranging from a few to several dozen yuan. Consequently, optimizing operational costs while maintaining service quality remains a critical challenge.

On-demand delivery becomes more and more popular due to its convenience and consumers' needs [4]. For the food delivery platform, the couriers can be mainly classified as dedicated couriers and crowdsourced couriers [5], and crowdsourced couriers are utilized to reduce the wait time of customers [6]. However, the voluntary order acceptance mechanism of crowdsourced couriers introduces significant variability in order fulfillment rates due to their order preferences [7]. Within algorithm-driven food delivery systems, couriers frequently suffer order assignment uncertainty as well as workload pressure [2]. This will further increase the uncertainty when accepting orders. Our field study of 51 couriers (information of this survey can be found in Appendix IV of Supplementary Material) reveals that nearly all crowdsourced couriers selectively reject unsuitable orders. This finding is corroborated

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The authors are with the SILC Business School, Shanghai University, Shanghai 201899, China (e-mail: sunxuting@shu.edu.cn; jchy2021@163.com; huojiajie@163.com).

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¹Online available at <https://www.statista.com/statistics/1235816/revenue-uber-eats-worldwide/> (accessed on Jan 2025).

²Online available at <https://www.statista.com/statistics/1235816/revenue-uber-eats-worldwide/> (accessed on Jan 2025).

³Online available at <https://www.businessofapps.com/data/food-delivery-app-report/> (accessed on Jan 2025).

⁴Online available at <https://www.chinabgao.com/k/zaixianwaimai/67455.html> (accessed on Jan 2025).

by our analysis of an instant delivery dataset,⁵ which shows that approximately 70% of couriers have rejected orders at least once. Notably, rejected orders may be reassigned up to 17 times, with an average reassignment duration of 118.85 seconds before final acceptance. From a sustainable operations perspective, incorporating courier preferences into dispatching decisions presents a dual benefit: it enhances system efficiency while aligning with platforms' broader social responsibility objectives.

Therefore, in this study, we focus on crowdsourced delivery problem in the context of on-demand food delivery, and incorporate the uncertainty of crowdsourced couriers' order preferences (i.e., acceptance probability) into the static scheduling model to optimize order assignment and routing decisions in a dispatch cycle. The specific workflow is as follows: The platform first collects orders and then assigns them to dedicated couriers and crowdsourced couriers. Crowdsourced couriers can choose to accept or reject orders, while dedicated couriers must accept assigned orders. We first predict the order acceptance probability of each crowdsourced courier and input the predicted values as a key parameter into the optimization model, thereby improving overall delivery system's efficiency. The main contributions are as follows:

(1) We identify eight features that influence crowdsourced courier order choices from a real delivery order dataset. We classify couriers using K-Means++ and build predictive models for each courier type to obtain acceptance probabilities. This guarantees that when dispatching orders, the platform offers couriers a more customized work experience.

(2) We develop an order assignment and routing model considering courier preferences to optimize order assignment strategies. To solve the model, we propose a logic-based Benders decomposition (LBBD) algorithm, and solves the original problem by decomposing it into order assignment and order sequencing. Also, we design an effective analytical cut to improve the algorithm's performance.

(3) Our model and algorithm can significantly reduce rejection rates and improve computational efficiency. In addition, sensitivity analysis reveals that the platform is in a dilemma in terms of operational costs and service quality. That is, when the platform is unwilling to re-dispatch orders frequently, i.e., increase the rejection coefficient, the proportion of orders assigned to dedicated couriers increases. However, this will lead to a surge in orders for dedicated couriers, which will negatively affect the delivery efficiency. Therefore, the platform has to assign orders to crowdsourced couriers, potentially increasing the dispatch times.

The rest of this paper is organized as follows. Section II reviews the relevant literature. Section III introduces the problem description and the mathematical formulation. Section IV provides the predictive model and the logic-based Benders decomposition algorithm. Section V shows the numerical results. Finally, Section VI concludes this paper.

II. LITERATURE REVIEW

In this section, we first conduct a comprehensive review of crowdsourced delivery problem. We particularly focus on the findings in areas such as compensation mechanisms and routing optimization. Then, we review the research works that are closely related to our work, namely the willingness to accept orders in the context of crowdsourced delivery.

A. Crowdsourcing Delivery

Crowdsourcing has attracted widespread attention due to its flexibility and cost-effectiveness. The Challenges and opportunities of crowdsourced delivery planning have been reviewed by Savelsbergh and Ulmer [8], [9]. Regarding compensation for crowdsourcing, Dahle et al. [10] introduced three compensation schemes and discussed their impact. In the first compensation scheme, the company provides a fixed and equal compensation for each service request, the second compensation is proportional to the travel cost from the pickup location to the delivery location, and in the third scheme, the compensation is proportional to the additional cost incurred by the crowdsourcing driver's detour. The results show that increasing the complexity of the compensation scheme will produce greater savings. Alnaggar et al. [11] provided a dynamic matching policy for the compensation problem of crowdsourcing drivers, which enabled crowdsourcing drivers to obtain a specific level of compensation while maintaining flexibility in their working hours. They proposed a Markov Decision Process (MDP) model to capture the dynamic and stochastic characteristics, and then designed a value function approximation algorithm to efficiently solve the MDP models.

Extensive research has been conducted on the crowdsourced delivery problem. With the emergence of crowdsourced logistics, considering the challenges due to increased uncertainty and dynamism, researchers have proposed dynamic routing models and stochastic routing models that incorporate stochastic demand or service time. Reyes et al. [12] introduced the dynamic meal delivery routing problem and developed a rolling-horizon repeated-matching algorithm to solve this problem in real time. Arslan et al. [13] innovatively investigated the routing problem involving both ad hoc drivers and dedicated vehicles. To address the matching between crowdsourcing and real-time tasks, they proposed a rolling horizon framework and developed an exact solution approach based on a matching formulation. Tao et al. [14] modeled the problem as a multi-warehouse pickup problem with dynamic drivers. They proposed an event-based rolling horizon approach, which updates the delivery route plan by iteratively calling a re-optimization process at each decision epoch. In the study by Silva et al. [15], not only the availability of crowdsourced drivers is uncertain, but also for the arrivals of orders. Therefore, they proposed a novel and exact solution method to compute the optimal prior solution before uncertainty is revealed. They then imposed probabilistic constraints to limit infeasible routes caused by capacity exceedance. Torres et al. [16] addressed the randomness of crowd drivers' availability by constructing a two-stage stochastic model and introduced a branch-and-price algorithm to solve routes. For large-scale

⁵Online available at <https://connect.informs.org/tsl/tslresources/datchallenge> (accessed on Jan2025).

problems, a column generation heuristic algorithm was also proposed. Côté et al. [17] proposed a two-stage formulation combining with a sample average approximation (SAA) method to address uncertainties in customer demand, location, and service time. In the first stage, delivery area time windows are allocated, while the second stage employs an adaptive large neighborhood search algorithm to optimize delivery routes.

B. Willingness to Accept Orders

There are a few studies in the literature that mention probability inference related to the willingness of crowdsourced drivers to participate in last mile delivery, as well as the potential impact of such participation on delivery costs. The existing literature mainly focuses on the factors that affect crowdsourcing couriers' acceptance behavior, while some studies focus on path planning. Hou [18] found that the age, income, and compensation per order are the most influential factors in the decisions of crowdsourced couriers to accept or reject delivery tasks through a questionnaire survey. Tao et al. [19] studied three settings for driver preferences: the minimum daily order quantity, the maximum order quantity allowed per trip, and the proportion of detours. The findings indicate that among all the design parameters of the O2O on-demand delivery system, the maximum number of orders allowed per trip and the detour ratio are crucial. Boysen et al. [20] investigated the impact of driver capacity, maximum acceptable detour, and minimum revenue per unit time on optimization models. They proposed an effective exact algorithm based on Benders decomposition.

Archetti et al. [21] explored the vehicle routing problem with occasional drivers, and the willingness of an occasional driver to deliver goods depends on whether the delivery location is too far from the driver's destination. Gdowska et al. [22] extended the research of Archetti et al. [21] by suggesting that drivers do not always accept tasks assigned to them. Therefore, they proposed a bi-level stochastic model. They introduced probability to represent the delivery willingness of each driver and solved the deterministic Vehicle Routing Problem after the tasks were accepted. Barbosa et al. [7] employed logistic regression to model the willingness of crowdsourced couriers to undertake deliveries. This model was utilized to establish a new compensation scheme based on the current routing plans of dedicated fleets and the probability of crowdsourcing acceptance. A direct search algorithm aimed at minimizing expected costs was adopted to determine the reward values for crowdsourcing. Ausseil et al. [23] proposed an uncertain dynamic matching problem regarding the acceptance or rejection of bids in crowdsourcing. They transformed the responses of crowdsourced drivers into probabilities of accepting future bids and utilized this information to optimize subsequent bidding decisions. The results indicate that online approximation and the incorporation of individual crowdsourced driver's acceptance estimates into order scheduling leads to a more efficient allocation. Çınar et al. [6] introduced the crowdshipping assignment problem with compensation-dependent acceptance probabilities. A mixed-integer nonlinear model for generic acceptance probability functions and an approximate linearization scheme was proposed. Additionally, an exact two-stage solution algorithm was

developed to separately handle compensation and assignment decisions. Some studies have employed machine learning to predict drivers' order acceptance behaviors. Wang et al. [11] used the extreme gradient boosting tree (XGBoost) model to predict crowdsourcing preferences in food delivery and designed an effective allocation sorting heuristic to generate order allocation schemes. They found that pickup distance, delivery cost, and remaining time are the main factors that couriers consider when selecting orders from the recommendation list, while delivery distance is not important. Tao et al. [2] introduced two machine learning-based models to predict the delivery speed and capacity of each crowdsourced driver in real-time, enabling better order allocation and routing decisions to minimize total costs. Similar research has emerged in the ridesharing domain. Horner et al. [24] developed a personalized menu to provide rideshare crowdsourced drivers with autonomy in their choices, gaining insights into drivers' request preferences through their expressed willingness to participate. The problem was addressed using the SAA method, and the results showed that the proposed stochastic optimization method outperformed current methods and deterministic optimization methods. Timilsina & Silvestri [25] investigated rideshare drivers' preferences for various tasks through spatial crowdsourcing, reinforcement learning and reverse auction theory. Based on the preferences, personalized lists are recommended to each driver. Then, drivers bid for their favorite tasks on their list through a reverse auction to achieve the optimal allocation of tasks. Yang et al. [26] conducted a probabilistic modeling of order acceptance and rejection behaviors among crowdsourced drivers to provide personalized delivery suggestion menus. They introduced two assortment optimization problems. For the disjoint problem, a standard greedy algorithm was proposed, while greedy and local search algorithms were employed for the joint assortment problem.

Although there are some studies that analyze the willingness of crowdsourced couriers to accept additional orders, the literature focusing on the on-demand food delivery process is scarce. Our research fills the gap regarding the modeling of delivery orders acceptance behavior of the crowdsourced couriers in the context of on-demand food delivery, which has not yet been systematically explored. Although machine learning has been widely applied to model and predict driver behavior, this has primarily been within ridesharing. However, in the context of on-demand food delivery, it lacks a deep exploration of the couriers' order acceptance behavior and its impact on the crowdsourcing delivery system. Thus, our study focuses on couriers' order acceptance behavior by analyzing various factors that affect their willingness to accept orders. In addition, we construct a novel order assignment and routing model to make a trade-off among operational cost, delivery efficiency, and couriers' welfare.

III. PROBLEM DESCRIPTION AND FORMULATION

A. Problem Description and Notations

In this section, an order assignment and routing problem in the context of instant food delivery is formulated, in which

both dedicated and crowdsourced couriers are considered. According to the real business operations [27], dedicated couriers are full-time employees of the delivery service platform, while crowdsourced couriers are part-time employees. Dedicated couriers cannot reject the orders dispatched by the dispatch system, whereas crowdsourced couriers can reject orders based on their preferences [6].

Specifically, whether a crowdsourced courier accepts the dispatched orders is uncertain and multiple factors affect the order acceptance behavior. According to Archetti et al. [21], the willingness of a crowdsourced courier to accept an additional order depends on the information of the current orders assigned to him/her. Besides, the information of the new order (e.g. meal preparation time, distance between the restaurant and customer) also affects the willingness of a crowdsourced courier to accept the order [1]. Due to different preferences of orders, different crowdsourced couriers will accept orders based on their own preference, and will show different order acceptance behaviors. The two kinds of historical information aforementioned are utilized to predict the order acceptance probability, which is incorporated into the crowdsourced delivery problem to consider the order acceptance behavior of each crowdsourced courier.

For the platform under consideration, the operating cycle has a length of T . Within the operating cycle $[0, T]$, orders are assigned to couriers with their routes in a time slot γ (e.g., 1 minute). Assuming T is a multiple of γ , the set of time period indices within the operating cycle is denoted by $\Gamma = \{1, \dots, \frac{T}{\gamma}\}$. In this study, we focus on the static order assignment and routing problem at a given dispatch cycle $t \in \Gamma$. At each dispatch cycle $t \in \Gamma$, there are two kinds of decisions to make: 1) Assignment decisions of the orders collected during the previous period $[\gamma(t-1), \gamma t]$ to the couriers; 2) Route replanning for each courier with his/her currently dispatched orders and the unfinished orders in the delivery process.

The instant delivery network can be represented by a network graph. Consider a directed graph $G = (V, A)$, where V is the set of all nodes and $A = \{(i, j) | i, j \in V, i \neq j\}$ is the set of all arcs between each pair of nodes. The set V include the pickup node set $P = \{1, \dots, n\}$, the delivery node set $C = \{n+1, \dots, 2n\}$, the courier set $K = \{1, \dots, k\}$, the set of pickup nodes for couriers' unfinished orders P^U , and the set of delivery nodes for couriers' unfinished orders C^U . An order i corresponds to a pickup node i and a delivery node $i+n$. The set of pickup nodes and delivery nodes is defined as $N = P \cup C$. The set of couriers K includes dedicated couriers K^D and crowdsourced couriers K^G . Each courier $k \in K$ has an dummy original node $o(k)$ and a dummy destination node $d(k)$. In each dispatch cycle, multiple orders can be assigned to each courier. According to the courier's experience level, there is a maximum number of orders that can be simultaneously served by courier k , denoted as L_k . The number of unfinished orders (i.e., orders on hand) of courier k is denoted as h_k .

The operational cost of the platform is composed of four parts. For dedicated couriers, they can get α reward for each order [19]. The reward of crowdsourced couriers is directly proportional to the delivery distance of orders, and can get β reward per unit kilometer [2]. Furthermore, each crowdsourced

TABLE I
NOTATION

Sets	Descriptions
P	Set of pickup nodes
C	Set of delivery nodes
N	Set of pickup and delivery nodes, i.e., $N = P \cup C$
C^G	Set of delivery nodes served by crowdsourced couriers
K^D	Set of dedicated couriers
K^G	Set of crowdsourced couriers
K	Set of couriers, i.e., $K = K^D \cup K^G$
C^U	Set of delivery nodes with unfinished orders for all couriers, where each order has been accepted by riders but has not yet been delivered
C_k^U	Set of delivery nodes with unfinished orders of courier k
P_k^U	Set of pickup nodes with unfinished orders by courier k
N_k^U	Set of pickup and delivery nodes with unfinished orders by courier k , i.e., $N_k^U = C_k^U \cup P_k^U$
Parameters	Descriptions
ρ_{ik}	Probability of courier k accepting order i
$o(k)$	Dummy original node of courier k
$d(k)$	Dummy destination node of courier k
b_i	Meal preparation completion time of order i
l_i	Latest delivery time of order i
L_k	The maximum number of orders served by courier k
d_{ij}	Distance between node i and node j
t_{ij}	Travel time between node i and node j
h_k	Number of unfinished orders for courier k
s_i	Service time of node i
M	A large number
α	Fees paid to dedicated couriers for each order
β	Fees paid to crowdsourced couriers per unit kilometer
c_d	Cost coefficient of unit overtime penalty
c_p	Cost coefficient of rejection probability
Variables	Descriptions
x_{ik}	Equal to 1 if order i is assigned to courier k , and 0 otherwise.
y_{ijk}	Equal to 1 if the courier k visits node i before node j , and 0 otherwise.
δ_i	The overtime duration of order i .
T_{ik}	Arrival time of the courier k at node i .

courier k accepts the order i with a probability ρ_{ik} , and the cost coefficient of rejection probability of crowdsourced couriers is c_p . Besides, each order has a meal preparation completion time b_i and a latest delivery time l_i . The service time of each node is s_i . The distance between any two nodes is defined as d_{ij} and the travel time of any two nodes is defined as t_{ij} . If the order is not delivered before the latest delivery time l_i , we set the penalty cost per unit time of overtime as c_d . The objective of the proposed optimization problem is to minimize the total costs, which consist of the labor costs of couriers, rejection cost and overtime cost. The main notations are shown in Table I.

Thus, the proposed problem is to determine the orders assigned to each courier and the sequence of the nodes to be visited by each courier so as to minimize the total costs for the platform. The preference of each crowdsourced courier ρ_{ik} is predicted through a data-driven approach (see section IV), which is further incorporated in the optimization model. Fig. 1 displays a feasible solution with 7 orders that are served by 2 dedicated couriers and 1 crowdsourced courier. The two dedicated couriers depart from the same dummy original node to complete Orders 1, 2, and Orders 3, 4 respectively. The crowdsourced courier needs to delivery Orders 5, 6, and 7.

Referring to the prior studies [1], [27], we make the following assumptions about the problem:

- 1) All couriers with electric bicycles have a constant and identical speed.

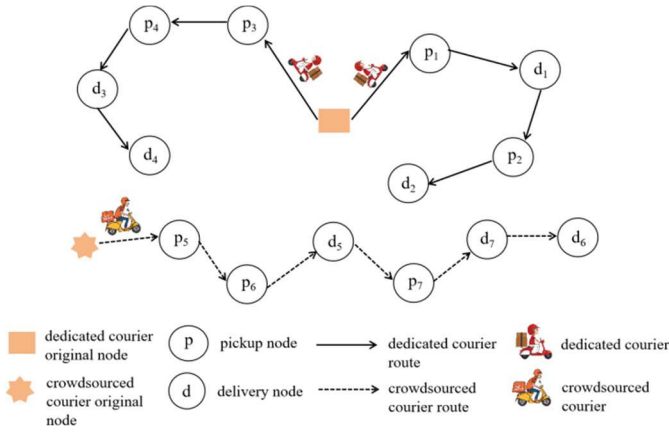


Fig. 1. An illustrative example for courier dispatching with both dedicated and crowdsourced couriers at a dispatch cycle t .

- 2) Order transfers between couriers after order assignment are not allowed.
- 3) Manhattan distance is used to calculate the distance between two nodes.
- 4) The individual order acceptance probability is independent from each other. When the platform dispatch more than one order to the courier at a time, we suppose the orders are listed in the order of the priori probability, which does not change the matching level between the courier and the order.

B. Mathematical Formulation

Based on the above assumptions and notation, the order assignment and routing model considering couriers' order preferences is constructed as follows. The objective function of the model (1) is to minimize the total cost of the platform. The first item represents the cost of the dedicated couriers, the second item represents the cost of the crowdsourced couriers, the third item represents the cost due to the potential crowdsourced couriers' orders rejection, and the fourth item represents the penalty cost due to overtime.

$$\min \sum_{k \in K^D} \sum_{i \in C} \alpha x_{ik} + \sum_{k \in K^G} \sum_{i \in C^G} \beta d_{i-n,i} x_{ik} + c_p \sum_{k \in K^G} \sum_{i \in C^G} (1 - \rho_{ik}) x_{ik} + c_d \sum_{i \in C \cup C^U} \delta_i \quad (1)$$

s.t.

$$\delta_i \geq T_{ik} - l_i, \forall k \in K, i \in C \cup C^U. \quad (2)$$

$$\sum_{k \in K} x_{ik} = 1, \forall i \in C^G. \quad (3)$$

$$\sum_{k \in K^D} x_{ik} = 1, \forall i \in C \setminus C^G. \quad (4)$$

$$x_{ik} = \sum_{j \in N \cup N_k^U \cup \{d(k)\}} y_{ijk}, \forall i \in C, k \in K. \quad (5)$$

$$\sum_{k \in K} \sum_{j \in N \cup \{o(k)\}} y_{jik} \leq 1, \forall i \in P \quad (6)$$

$$\sum_{k \in K} \sum_{j \in N \cup \{d(k)\}} y_{ijk} \leq 1, \forall i \in C. \quad (7)$$

$$\sum_{j \in N \cup N_k^U \cup \{o(k)\}} y_{jik} = 1, \forall k \in K, i \in P_k^U. \quad (8)$$

$$\sum_{j \in N \cup N_k^U \cup \{d(k)\}} y_{ijk} = 1, \forall k \in K, i \in C_k^U. \quad (9)$$

$$\sum_{j \in P \cup P_k^U \cup \{d(k)\}} y_{o(k)jk} = \sum_{j \in C \cup C_k^U \cup \{o(k)\}} y_{jd(k)k} = 1, \forall k \in K. \quad (10)$$

$$\sum_{j \in N \cup N_k^U \cup \{d(k)\}} y_{ijk} - \sum_{j \in N \cup N_k^U \cup \{o(k)\}} y_{jik} = 0, \quad \forall i \in N \cup N_k^U, k \in K. \quad (11)$$

$$\sum_{j \in N \cup N_k^U} y_{ijk} = \sum_{j \in N \cup N_k^U} y_{j,n+i,k}, \quad \forall k \in K, i \in P \cup P_k^U. \quad (12)$$

$$T_{ik} + s_i + t_{ij} \leq T_{jk} + M(1 - y_{ijk}), \quad \forall i, j \in N \cup N_k^U \cup \{o(k), d(k)\}, k \in K. \quad (13)$$

$$T_{ik} + s_i + t_{i,n+i} \leq T_{n+i,k}, \forall i \in P \cup P_k^U, k \in K. \quad (14)$$

$$T_{ik} \geq b_i, \forall i \in P \cup P_k^U, k \in K. \quad (15)$$

$$h_k + \sum_{i \in C} x_{ik} \leq L_k, \forall k \in K. \quad (16)$$

$$x_{ik} \in \{0, 1\}, \forall i \in C, k \in K. \quad (17)$$

$$y_{ijk} \in \{0, 1\}, \forall i, j \in N \cup N_k^U \cup \{o(k), d(k)\}, i \neq j, k \in K. \quad (18)$$

$$T_{ik} \geq 0, \forall i \in N \cup N_k^U \cup \{o(k), d(k)\}, k \in K \quad (19)$$

$$\delta_i \geq 0, \forall i \in C \cup C^U. \quad (20)$$

Constraint (2) defines the overtime for each delivery node. Constraints (3) and (4) indicate that each order is assigned to a courier, where dedicated couriers can serve all orders, while crowdsourced couriers can only serve a subset of orders. Constraint (5) means that delivery nodes assigned to each courier are directly served by the courier. Constraints (6) and (7) indicate that each pickup node and delivery node can only be visited by one courier once. Constraints (8) and (9) specify that unfinished orders for couriers must be visited by the couriers. Constraint (10) means that the dedicated couriers and crowdsourced couriers need to start from the original node and return to the destination node. Constraint (11) ensures the flow balance. Constraint (12) dictates that the same courier must visit both the pickup and delivery nodes for the same order. Constraint (13) states that if the courier visits the node j immediately after the node i , the arrival time at the node j must be greater than or equal to the sum of the arrival time at node i , the service time at node i and travel time between the two nodes. Constraint (14) indicates that couriers must first visit the pickup node before visiting the delivery node. Constraint (15) states that the time when the courier arrives at the pickup node cannot be earlier than the meal preparation time of the order. Constraint (16) specifies the maximum order capacity for a courier. Constraints (17) - (20) are variables' domain constraints.

IV. SOLUTION METHODS

In this section, we propose a data-driven two-module framework that synergistically integrates machine learning and operations research techniques for the order assignment and routing problem. As illustrated in Fig. 2, in the predictive module, we employ XGBoost to predict the acceptance probability of each order for clustered crowdsourced couriers. In the optimization module, these predicted probabilities are then

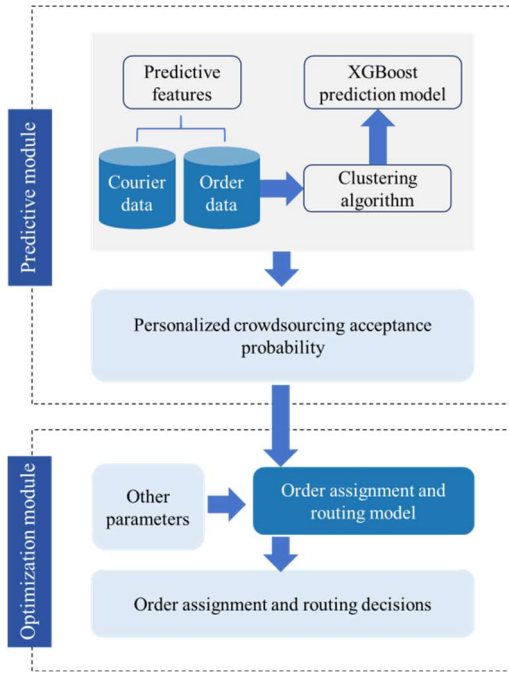


Fig. 2. The data-driven two-module framework.

incorporated into the optimization model, which is solved using a logic-based Benders decomposition algorithm.

A. Predictive Model

In this section, a predictive model for crowdsourced couriers' order acceptance probability is developed. First, we figure out eight features based on the dataset as preliminary work for crowdsourced couriers' clustering. After characterizing the crowdsourced couriers into groups, we apply four predictive machine learning algorithms to predict the individual order acceptance probability of each crowdsourced courier. The best predictive algorithm is selected after performance evaluation.

1) *Feature Engineering*: In this section, we conduct feature engineering to figure out the crowdsourced couriers' order preferences and characterize them into different groups. According to the dataset, we design two categories (i.e. courier features and order features) with a total of eight features to describe crowdsourced couriers' order acceptance behavior, as shown in the Table II.

The methodology to quantify the courier features are described as below. To determine the orders on hand (R1), firstly, we identify all orders under the current courier ID. Then, we filter out the orders whose dispatch time is earlier than the current dispatch time and the delivery time is later than the current dispatch time as R1.

For the following four features (i.e., R2, R3, R4 and R5), the first step is to obtain the restaurant locations and customer locations corresponding to the courier's orders on hand (i.e., unfinished orders). By identifying the orders on hand of the current courier ID, the corresponding restaurant locations can be obtained from the restaurant's latitude and longitude accordingly. Similarly, the corresponding customer locations can be obtained from the customer's latitude and longitude. Afterwards, we calculate the maximum distance

TABLE II
DESCRIPTION OF FEATURES

Feature category	Feature	Description
Courier features	Orders on hand (R1)	The number of orders previously assigned to a courier that have not yet been finished
	Max distance from current restaurant to unfinished restaurants (R2)	The maximum distance from the current order's restaurant r to any other restaurant with unfinished orders R^U , i.e. $\max_{r' \in R^U} d(r, r')$
	Max distance from current customer to unfinished restaurants (R3)	The maximum distance from the current order's customer c to any other restaurant with unfinished orders R^U , i.e. $\max_{r' \in R^U} d(c, r')$
	Max distance from current restaurant to unfinished customers (R4)	The maximum distance from the current order's restaurant r to any other customer with unfinished orders C^U , i.e. $\max_{c' \in C^U} d(r, c')$
	Max distance from current customer to unfinished customers (R5)	The maximum distance from the current order's customer c to any other customer with unfinished orders C^U , i.e. $\max_{c' \in C^U} d(c, c')$
Order features	Remaining delivery time (O1)	The remaining delivery time of the order at the time of current dispatch
	Meal preparation time (O2)	The preparation time of the restaurant
	Restaurant customer distance (O3)	The distance between a restaurant and a customer corresponding to an order

from the restaurant location of the current dispatched order to those restaurant locations of the courier's unfinished orders (R2). Similarly, we can figure out the maximum distance from the customer location of the current dispatched order to all restaurant locations of the courier's unfinished orders (R3); the maximum distance from the restaurant location of the current dispatched order to all customer locations of the courier's unfinished orders (R4); the maximum distance from the customer location of the current dispatched order to all customer locations of the courier's unfinished orders (R5).

The methodology to quantify order features are shown below. O1 is obtained by subtracting the dispatch time from the estimated arrival time. O2 is obtained by subtracting the platform order time from the estimated meal preparation time. O3 is calculated based on the coordinates of the restaurant and the customer, representing the Manhattan distance in a two-dimensional plane.

Finally, outliers for the above eight characteristics are identified and removed using the interquartile range method, resulting in 4,881 couriers with a total of 298,211 data. An outlier is the value which is statistical anomaly and significantly deviates from other observations.

2) *Crowdsourced Couriers' Order Preferences Clustering*: Due to the limited data for some crowdsourced couriers, building a predictive model for each courier would result in underfitting during the training process and lower prediction accuracy. Hence, before predicting the individual probability for each crowdsourced courier, we use the K-Means++ clustering method to characterize the crowdsourced couriers into groups. By clustering couriers with similar behavioral characteristics into groups for model training, the issue of insufficient courier data is addressed. For the selection of initial cluster centers, the K-means++ algorithm improves on the K-means algorithm's random selection by employing a probabilistic method. This algorithm ensures that the distances

between cluster centers are as far as possible, reducing the risk of getting stuck in local optimal solutions and achieving better clustering results and faster convergence speed.

Algorithm 1 The framework of predictive model based on K-Means++

Input Π (eight feature sets), L_{max} (number of clusters), I_{max} (number of iterations)

Output Best number of clusters L^* , $best_{clusters}$, $best_{model}$

```

1 For  $L = 2$  to  $L_{max}$  do
2   Initialize cluster centers through K-Means++
3   For  $i = 1$  to  $I_{max}$  do
4     Initialize cluster labels, i.e., the index of the cluster center
5     Calculate the distance  $dist$  between the features of each data point and the cluster center
6     Select the cluster center with the smallest  $dist$  and update the clustering label
7     Until each data point is assigned a clustering label
8     Recalculate the cluster center of the same cluster label set and update the cluster center
9   End for
10  Predict in each cluster set
11  Calculate the AverageAUC of all predictive models for the number of clusters  $L$ 
12 End for
13 Return the optimal number of clusters  $L^*$ , optimal clustering result  $best_{clusters}$  and optimal clustering model  $best_{model}$  that maximize the Average AUC

```

The clustering and prediction processes are shown in Algorithm 1. The inputs are the features of the crowdsourced couriers, the maximum number of clusters L_{max} and the maximum number of iterations I_{max} . In order to set a reasonable upper limit L_{max} for the number of clusters, we conduct preliminary experiments. We find that when L_{max} is set to 15 or above, the clustering effect does not improve, but the model's training time increases a lot. Therefore, we set parameter L_{max} as 15. Finally, the model outputs the best number of clusters L^* , best clustering result $best_{clusters}$ and best clustering model $best_{model}$.

3) *Predictive Algorithms*: To further predict the individual order acceptance probability of each crowdsourced courier who belongs to a specific cluster figured out in the previous subsection, in this section, we introduce four predictive methods, including logistic regression (LR), random forest (RF), XGBoost, and support vector machine (SVM) [18], [28]. The explanation on how the four methods are applied to predict the order acceptance probability is shown in Appendix VI of Supplementary Material. We evaluate the performance of those four machine learning techniques and choose the best one for individual probability prediction.

B. Logic-Based Benders Decomposition Algorithm

The proposed problem consists the assignment of orders to couriers and the sequencing of orders for each courier. The two subproblems are interconnected, and the optimization method to solve the proposed problem should deal with the subproblems properly [29]. The fundamental concept of Benders

decomposition is to decompose the original problem into a relaxed master problem and a linear subproblem [30]. After the master problem passes the optimal solution to the subproblem, the subproblem is solved continuously to generate Benders cuts, which are added to the master problem. Furthermore, the subproblem in this study can be further decomposed into subproblems for each courier. Thus, the master problem and the subproblem can interact iteratively, and the computational burden is expected to be reduced. In our problem, the classical Benders decomposition is not applicable because it requires the subproblem to be a continuous linear programming problem. However, our subproblem involves an order sequencing decision problem that includes integer decision variables. Therefore, this paper employs a LBBD algorithm to solve the subproblem with integer variables [31]. The LBBD derives Benders cuts by solving a more general reasoning dual, rather than applying linear programming duality to the subproblem as in classical Benders decomposition. According to the problem features of our study, the original problem is divided into a master problem for order assignment and a subproblem for order sequencing.

1) *Master Problem*: We introduce a continuous variable σ to represent the lower bound of the overtime penalty cost of all orders. Therefore, the master problem can be expressed as follows.

$$\min \sum_{k \in K^D} \sum_{i \in C} \alpha x_{ik} + \sum_{k \in K^G} \sum_{i \in C^G} \beta d_{i-n,i} x_{ik} + c_p \sum_{k \in K^G} \sum_{i \in C^G} (1 - \rho_{ik}) x_{ik} + \sigma \quad (21)$$

$$\text{s.t. (3), (4), (16) - (18), and} \\ [Benderscuts] \quad (22)$$

2) *Strengthen the MP*: To obtain a tighter lower bound σ for the master problem, the following inequality (23)-(25) is employed to strengthen the master problem. First, we introduce the continuous variable ξ_k to represent the total overtime penalty cost of the orders assigned to courier k . Then, the lower bound of the overtime penalty cost ξ_k can be expressed in inequality (24), in which the first term represents the minimum overtime penalty cost incurred by the courier when new orders are assigned, and the second term represents the minimum overtime penalty cost arising from the on-hand orders of the courier.

$$\xi_k \geq 0, \forall k \in K. \quad (23)$$

$$\xi_k \geq c_d \sum_{i \in C} x_{ik} (t_{k,i-n} + s_{i-n} + t_{i-n,i} - l_i) + c_d \sum_{i \in C_k^U} \delta_i, \quad \forall k \in K. \quad (24)$$

$$\sigma \geq \sum_{k \in K} \xi_k. \quad (25)$$

3) *Sequencing Subproblem*: The subproblem is to determine the delivery sequence for each courier. For each courier k , there are two sets of orders: 1) the orders dispatched to him/her for the current dispatch cycle $C_k = \{i | \hat{x}_{ik} = 1, i \in C\}$. This is determined by the master problem; 2) the unfinished orders, which are the state of each courier known at the beginning of the dispatch cycle C_k^U . P_k , corresponding to C_k , refers to the set of pickup nodes visited by the courier, and $N_k = P_k \cup C_k$ refers

to the set of both pickup and customer nodes visited by the courier. Similarly, P_k^U corresponds to C_k^U , and $N_k^U = P_k^U \cup C_k^U$. For each courier k , it is an independent routing problem. The binary decision variable y_{ij} ($i \neq j; i, j \in C_k$) is introduced, which means that if the courier k visits node i first and then node j , it is 1, otherwise it is 0. Thus, the subproblem is as follows.

$$\min c_d \sum_{i \in C_k \cup C_k^U} \delta_i \quad (26)$$

s.t.

$$\delta_i \geq T_i - l_i, \forall i \in C_k \cup C_k^U. \quad (27)$$

$$\sum_{j \in N_k \cup N_k^U \cup \{d(k)\}} y_{ij} = 1, \forall i \in N_k. \quad (28)$$

$$\sum_{j \in P_k \cup P_k^U \cup \{d(k)\}} y_{o(k)j} = \sum_{j \in C_k \cup C_k^U \cup \{o(k)\}} y_{jd(k)} = 1. \quad (29)$$

$$\sum_{j \in N_k \cup N_k^U \cup \{d(k)\}} y_{ij} - \sum_{j \in N_k \cup N_k^U \cup \{o(k)\}} y_{ji} = 0, \forall i \in N_k \cup N_k^U. \quad (30)$$

$$T_i + s_i + t_{ij} \leq T_j + M(1 - y_{ij}), \quad \forall i, j \in N_k \cup N_k^U \cup \{o(k), d(k)\}. \quad (31)$$

$$T_i + s_i + t_{i,n+i} \leq T_{n+i}, \forall i \in P_k \cup P_k^U. \quad (32)$$

$$T_i \geq b_i, \forall i \in P_k \cup P_k^U. \quad (33)$$

$$\delta_i \geq 0, \forall i \in C_k \cup C_k^U. \quad (34)$$

$$y_{ij} \in \{0, 1\}, \forall i, j \in N_k \cup N_k^U \cup \{o(k), d(k)\}, i \neq j. \quad (35)$$

$$T_i \geq 0, \forall i \in N_k \cup N_k^U. \quad (36)$$

4) *Optimality Cut*: After the subproblems are solved, Benders cuts are generated and added to the master problem. In our formulation, the solution of the master problem is always feasible for the subproblems. Therefore, we only generate optimality cuts to enhance the lower bound of the objective function of the master problem. Let C_k^h be the customers set served by the courier k at iteration h , and $\xi_k^h(C_k^h)$ be the overtime cost of the courier k at iteration h . For any courier, the optimality cuts are shown in inequality (37). This inequality ensures that if the set of customers served by courier k in $h+1$ th iteration includes C_k^h , then ξ_k^{h+1} is not lower than $\xi_k^h(C_k^h)$. However, once a customer is removed from C_k^h , the cuts will no longer be binding. Since $\xi_k^h(C_k^h) \left(1 - \sum_{i \in C_k^h} (1 - x_{ik})\right)$ is non-positive, this inequality must hold [32], [33].

$$\xi_k^{h+1} \geq \xi_k^h(C_k^h) \left(1 - \sum_{i \in C_k^h} (1 - x_{ik})\right), \forall k \in K. \quad (37)$$

$$\xi_k^h(C_k^h) = c_d \sum_{i \in C_k^h} \delta_i, \forall k \in K. \quad (38)$$

Next, following the existing literature [34], [35], we introduce a method to strengthen this inequality, that is, to find the minimal subset C_k^{h*} of C_k^h . We iteratively compute the minimum overtime cost $\xi_k^h(C_k^h \setminus \{j\})$ for removing a customer node j from C_k^h . If $\xi_k^h(C_k^h \setminus \{j\})$ is equal to $\xi_k^h(C_k^h)$, we remove the customer node j from C_k^h until all customer nodes have been enumerated. Let R_k^h denote the set of customers that can

be removed from C_k^h , i.e. $R_k^h = \{j \in C_k^h | \xi_k^h(C_k^h \setminus \{j\}) = \xi_k^h(C_k^h)\}$. $\xi_k^h(C_k^{h*})$ represents the overtime cost after removing R_k^h customers from C_k^h , i.e. $C_k^{h*} = C_k^h \setminus R_k^h$, so there is the following inequality.

$$\xi_k \geq \xi_k^h(C_k^{h*}) \left(1 - \sum_{i \in C_k^{h*}} (1 - x_{ik})\right), \forall k \in K. \quad (39)$$

In subsequent iterations, the strengthened inequality is only valid when the values of the decision variables in set C_k^{h*} are identical to those in the current iteration, specifically when $x_{ik} = 1, \forall i \in C_k^{h*}$. To provide more information to the master problem, customers in set C_k^h are no longer assigned to k in subsequent iterations, and customers outside of set C_k^h can be assigned to k . Based on the characteristics of our problem, we propose stronger cuts (40), namely analytical Benders cuts [34]. For this inequality, if customer i is no longer assigned to the courier k in subsequent iterations, the overtime cost can be reduced by $c_d(2t_{i-n}^{max} + s_{i-n} + t_i^{max} - t_{kh}^{min})^+$ at most. Therefore, even if customers' assignments change, this inequality still offers a lower bound for the overtime cost. See the next subsection for the proof of this inequality.

$$\xi_k \geq \xi_k^h(C_k^h) - \sum_{i \in C_k^h} c_d(1 - x_{ik}) (2t_{i-n}^{max} + s_{i-n} + t_i^{max} - t_{kh}^{min})^+, \quad \forall k \in K \quad (40)$$

In inequality (40), t_{kh}^{min} represents the shortest travel time along the edges associated with the nodes in set $C_k^h \cup P_k^h \cup C_k^U \cup \{k\}$, that is, $t_{kh}^{min} = \min\{t_{ij} | i, j \in C_k^h \cup P_k^h \cup C_k^U \cup \{k\}\}$. t_i^{max} is the longest time from node $i \in N$ to other nodes, that is, $t_i^{max} = t_{ij}$.

5) *Validity of the Proposed Benders Cut*: This section demonstrates the effectiveness of the proposed analytical Benders cuts. Referring to the existing literature [36], [37], an effective cut must satisfy two conditions: (i) it must eliminate suboptimal solutions generated by the current master problem; (ii) it cannot eliminate any global optimal solutions. The first condition is straightforward. If the same set of orders is assigned to the same courier in subsequent iterations, the variable value of the overtime cost in the master problem should at least become worse, so that the cut can effectively remove the current solution. Otherwise, the assignment to the courier must be altered. In any case, the solution at iteration h will be excluded from the master problem space. The second condition is somewhat more complex, and we provide the following proof.

Theorem 1: Given iteration h , if the customer set assigned to courier k is C_k^h , the minimum overtime cost is $\xi_k^h(C_k^h)$. If any customer i is removed from set C_k^h in subsequent iterations, the reduction of the minimum overtime cost of the courier will not exceed $c_d \max\{0, 2t_{i-n}^{max} + s_{i-n} + t_i^{max} - t_{kh}^{min}\}$.

Proof. Assuming that in iteration f ($f > h$), the set of customers assigned to courier k is $C_k^h \setminus \{i\}$, which means that customer i is removed from C_k^h . Consequently, the optimal path for iteration f is derived from iteration h by deleting three edges and adding one edge, as illustrated in Fig. 3. This figure depicts the routes of two couriers after they have removed customers 2 and 4 respectively in iteration h . The black line represents the optimal path of iteration h , the blue dotted line

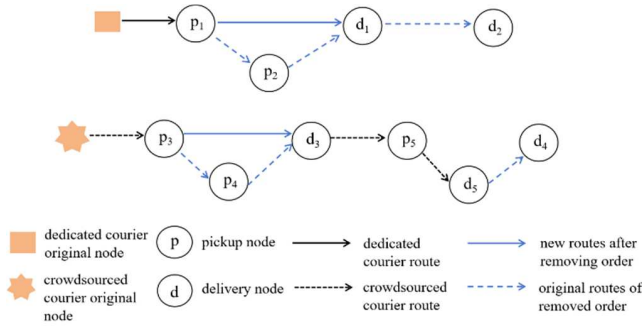


Fig. 3. Routes after removing orders.

represents the edge removed in iteration f , and the blue solid line represents the edge added in iteration f . It can be seen that removing a customer i will remove the edge entering the customer and the two edges of the pickup connection corresponding to the customer. The maximum reduction of the edge connected with pickup $i - n$ is t_{i-n}^{max} , the maximum reduction of the edge connected with customer i is t_i^{max} , and the minimum increase of the regenerated edge is t_{kh}^{min} . Therefore, after removing a customer, the minimum overtime cost will not exceed $c_d \max\{0, 2t_{i-n}^{max} + s_{i-n} + t_i^{max} - t_{kh}^{min}\}$.

6) *Dynamic Programming Algorithm for Subproblems*: To minimize overtime costs, the subproblem deals with order sequencing for each courier. It is found that solving the subproblem is very time-consuming compared to the master problem, which significantly reduces the overall efficiency of the LBB. Therefore, a specific dynamic programming algorithm is proposed to solve the sorting problem of orders for each courier, which improves algorithm efficiency.

Algorithm 2 Dynamic programming for solving subproblems

Input Pickup nodes, customer nodes, couriers' original position, meal preparation time and latest delivery time of orders, service time, and travel time matrix

Output Optimal route and minimum overtime duration

- 1 Define the mapping function of the nodes
- 2 Initialize dp (dynamic programming table) $\leftarrow \infty$ and $parent(predecessor\ node) \leftarrow -1$
- 3 **For** set of all possible accessed points S :
- 4 **For** all possible current nodes i :
- 5 **If** current status is valid (i.e., $dp \neq \infty$):
- 6 **For** unvisited node j :
- 7 **If** node j is a delivery node and the corresponding pickup node is not accessed:
- 8 **continue**
- 9 **Else**:
- 10 Calculate the minimum overtime according to the recursive equation (41)
- 11 Update dp and $parent$
- 12 **End for**
- 13 **Return** optimal route and minimum overtime

The recursive equation is shown in equation (41), where $f(C, *)$ represents the minimum overtime when the order set is C , t_j represents actual arrival time of the order j , and l_j represents the latest delivery time of the order j . This recursive

equation is initialized to $f(\emptyset, *) = 0$, and the algorithm framework of dynamic programming is shown in Algorithm 2 below. First, a mapping function between the original point set and the newly numbered point set is established. Then, recursive equation (41) is employed to calculate the shortest arrival times and overtime from the original point to other points. The algorithm gradually updates the arrival time and overtime through the dynamic programming table (dp), ensuring the constraints of pickup time windows and pickup priority are satisfied. Finally, the optimal path and overtime are obtained by backtracking through parent.

$$f(C, *) = \min \{f(C \setminus \{j\}, *) + \max\{t_j - l_j, 0\}\} \quad (41)$$

7) *Branch-and-Check*: The classical LBB method solves the master problem to optimality first, followed by solving subproblems to generate Benders cuts, which are then added to the master problem for iterative solving. In reality, it has been observed that the time taken to solve the master problem to optimality increases as the continuous addition of Benders cuts. In this section, we introduce the branch-and-check framework which is an exact algorithm technique to enhance the effectiveness and efficiency of solving the proposed MILP models. For solving complex master problems and simple subproblems, it has been proven that branch-and-check is an efficient method [37]. In branch-and-check, the master problem is solved using the branch-and-cut method and is solved only once. Each time the feasible solution of the master problem is found, the subproblem is then solved. Benders cuts are generated and added to each node of the master problem search tree. Define z_k as the upper bound found by the subproblem in iteration k . If the lower bound v_k found by the master problem is equal to z_k , then a global optimal solution is found. Otherwise, Benders cuts generated from subproblems are added to the master problem to continue solving until the globally feasible solution is optimal. The algorithm framework of the whole LBB is shown in Algorithm 3.

Algorithm 3 LBB for the order assignment and routing problem

- 1 Initialization $Cuts \leftarrow \emptyset, z_{min} \leftarrow \infty$
- 2 **While** $v_k \neq z_{min}$ and the time limit is not exceeded **do**
- 3 Solve the master problem of order assignment in a single search tree branch
- 4 **If** a feasible solution x_k , objective value v_k **do**
- 5 Dynamic programming to solve the subproblems given x_k , the optimal value z_k
- 6 **If** $z_k < z_{min}$ **then**
- 7 Update upper bound $z_{min} \leftarrow z_k$
- 8 **End**
- 9 Generate Benders cuts
- 10 Update cut and add cuts to other nodes of the search tree of the master problem
- 11 Repeat 2-10
- 12 **End while**
- 13 **Return** Optimal solution and optimal objective value

TABLE III
DATA CHARACTERISTICS AND DESCRIPTION

Field Name	Description	Example
Order id	Unique and anonymized identifier of an order	12
Waybill id	Unique and anonymized identifier of a waybill. An order may have several waybill IDs; a waybill is created when an order enters the dispatch system for assignment; a new waybill is created when the order is refused by a courier and re-enters the assignment process	2
Courier id	Unique and anonymized identifier of the courier	4
Da id	Unique and anonymized identifier of the business district	0
Is courier grabbed	Whether the waybill is accepted by the courier	1
Poi id	Unique and anonymized identifier of the restaurant	0
Is weekend	Whether the order date is weekend	0
Sender lng	Longitude of the pickup point	174579111
Sender lat	Latitude of the pickup point	45862631
Recipient lng	Longitude of the delivery point	174600135
Recipient lat	Latitude of the delivery point	45852786
Dispatch time	Unix time when a waybill is assigned to a courier	1665936258
Grab time	Unix time when the waybill is accepted by the courier	174530062
Fetch time	Unix time when the waybill is fetched by the courier	1665937344
Platform order time	Unix time when the order is created	1665935996
Estimate meal prepare time	Unix time when meal preparation is complete	1665936779
Estimate arrived time	Unix time when the order is promised to be delivered to the customer	1665937864
Arrive time	Unix time when the waybill is delivered by the courier	1665937798

V. COMPUTATIONAL EXPERIMENTS

In order to evaluate the performance of the proposed algorithm, numerical experiments are conducted using real data from Meituan to generate instances of various scales. All models and algorithms are coded in Python 3.7 linked with GUROBI 11.0.2. All the computational experiments are conducted on a personal computer equipped with Intel(R) Core(TM) i9-14900HX CPU (2.20 GHz) and 32 GB RAM. The running time limit for all instances is set to 3600 seconds.

A. Data and Instance Generation

1) *Data for the Predictive Model*: The data comes from the data-driven research competition jointly organized by Informs and Meituan.⁶ This data contains 654,343 order information from Meituan between October 17, 2022 and October 24, 2022, involving 4954 couriers. The fields included in the data are shown in Table III below.

The original data often has problems with data quality, which are mainly shown in three aspects: (1) incompleteness, where important fields have missing values; (2) inconsistency, which is defined by abnormal values; and (3) redundancy, which involves duplicate records. Our study applies a standardized procedure to the data, which includes data cleaning and data transformation, in order to improve data quality and the reliability of the analytical results.

Data cleaning involved addressing outliers and missing values in several fields. For the “platform order time” field,

anomalous records with a value of 0 are removed. For time-related fields, such as “estimate meal prepare time”, records with null values are eliminated. The final cleaned dataset comprises 622,757 valid records from 4,938 riders, with a data completeness rate of 95.17%.

Data transformation included the normalization of geographic coordinates and time fields. For the latitude and longitude fields, a two-step conversion is applied: (1) Format conversion: Integer-stored coordinates (e.g., 174579111, 45862631) are converted to decimal format (e.g., 174.579111, 45.862631); (2) Spatial mapping: Coordinates are projected onto a two-dimensional plane using the Web Mercator projection. For time-related fields, standardization involves converting Unix timestamps into a standard date-time format using the `fromtimestamp` function from Python’s `datetime` module. For instance, the timestamp 1665977925 is transformed into the format of “2022-10-17 11:38:45”.

2) *Instance Generation for the Optimization Model*: To obtain the optimal solutions in a reasonable time, and conduct comparisons between Gurobi and the proposed LBBP regarding the solution quality, we test small to medium size instances (i.e., $n = 10, 20, 30, 40$). In addition, we extract the order information from the real dataset in the same time slot (i.e., 1 minute) to form an instance. For example, we extract 10 orders’ information in the same time slot to form an instance with $n = 10$. The number of couriers is determined by the information in the dataset, where the number of dedicated couriers and the number of crowdsourced couriers are set to be 1:1. Five instances are generated for each scale. For each instance, it is assumed that crowdsourced couriers can only serve 50% of customers, while dedicated couriers can serve all customers. The parameters for each instance are set as follows.

First, the longitude and latitude coordinates of merchants, customers, and couriers in the dataset are transformed into two-dimensional plane coordinates. Then we calculate the distance of any two coordinates according to the Manhattan distance. According to the speed limit of electric bicycle (which is the main transport tool of couriers), we set the courier’s driving speed as 25 km/h.⁷ Besides, according to the feedback from the couriers of Meituan and Ele.me platforms, the delivery fee per order is around 4-6 RMB. The delivery range is usually less than 3km. Therefore, we set the delivery fee per order for a dedicated courier is 5 RMB. For a crowdsourced couriers, the delivery fee per km is 1.5 RMB. Similarly, we set the unit overtime penalty cost and the cost coefficient of the rejection probability as 0.6 and 0.5, respectively. We also conduct sensitivity analyses in terms of different parameters such as delivery fee, penalty, to check the robustness of the model’s performance in Section V PART D. The meal preparation time and expected delivery time are obtained from the variables “estimate meal prepare time” and “estimate arrived time” respectively. The weight of each order is randomly generated from the uniform distribution in $U[1, 5]$. The service time of each node is randomly generated from the uniform distribution in $U[10, 240]$. The order capacity of

⁶Online available at <https://connect.informs.org/tsl/tslresources/datchallenge> (accessed on Jan2025).

⁷Online available at https://www.gov.cn/zhengce/202501/content_6998231.htm (accessed on June 2025).

TABLE IV
RESULTS COMPARISON OF FOUR PREDICTIVE MODELS

Model	Best_L	Average AUC	Average Recall	Run time
LR	12	0.6273	0.6253	33123 s
RF	6	0.7121	0.6345	37855 s
XGBoost	6	0.7446	0.8689	33695 s
SVM	8	0.6939	0.7001	54350 s

couriers is randomly generated from the uniform distribution in $U[2, 13]$ based on the existing orders on hand.

B. Results for the Predictive Model

In this section, we compare the differences in predictive performance among four methods. In addition, we also introduce the tuning of XGBoost hyperparameters and explain the model through feature importance ranking.

1) *Performance Comparison of Predictive Models*: The predictive results of the four models are shown in Table IV. Best_L represents the optimal number of clusters. Average AUC represents the average Area Under Curve of all clusters, and is used to evaluate the performance of the classification model. The range of AUC values is from 0 to 1, with larger values indicating stronger classification ability. Average Recall refers to the average recall rate of all clusters, which measures the model's ability to identify positive samples. Run time represents the running time of the model. In the comparison of four predictive models, it is found that XGBoost performed the best, with the highest Average AUC (0.7446) and Average Recall (0.8689). This indicates that 86.89% of the samples that actually accepted are correctly predicted. And the AUC of RF (0.7121) is second to XGBoost (0.7446). LR has the shortest training time (33123s) and is suitable for quickly obtaining model results, but its AUC (0.6273) and Recall (0.6253) are relatively low. Although SVM has some classification ability, its training time is the longest. SVM performs poorly in both performance and computational efficiency. Overall, XGBoost is the optimal choice. The XGBoost predictive model is optimized through hyperparameter tuning, and the optimal values of XGBoost hyperparameters are shown in Table AI in Appendix I of Supplementary Material.

2) *Interpretability of the Model*: Feature importance analysis provides a perspective for understanding the interpretability of models. By analyzing feature importance, we can identify which factors play a key role in the decision-making process. In our study, the importance ranking of each feature in the six categories derived from XGBoost is shown in Fig. A1 in Appendix II of Supplementary Material. At the same time, we aggregate the daily order acceptance rate of each courier type aimed to analyze the relationship between the couriers' preferences and their work experience. The aggregated data are presented in Table AII in Appendix II of Supplementary Material.

We can observe that the sixth category of couriers (Fig. A1f) has the lowest average daily order acceptance rate, while the first five categories of couriers (Fig. A1a-1e) have daily order acceptance rates above 80%. Consequently, we initially categorize the six types of couriers into two major groups

based on their job experience. For Fig. A1f, this group consists of couriers with little job experience, for whom delivery distance is a key factor. These couriers are unfamiliar with the delivery environment and are more likely to accept small-scale orders with concentrated geographical distribution.

For the first five categories of couriers, the order acceptance rate in Fig. A1a is relatively low. This category of couriers demonstrates high autonomy and flexibility in order acceptance. They pay more attention to timeliness and workload, tending to optimize the efficiency of current tasks.

Figure A1(b) depicts couriers who are extremely time-sensitive. The remaining delivery time (O1) carries the highest weight, followed by the number of orders on hand (R1). This indicates that such couriers are highly sensitive to overtime risks while also being mindful of their workload. They tend to complete existing orders before accepting new ones.

Figure A1(c) illustrates dynamically coordinated couriers. While the remaining delivery time (O1) and the number of orders on hand (R1) remain dominant, the spatial feature weights (R2, R3, R4, R5) are relatively balanced. This results in a decision-making process that balances time, task load, and spatial dimensions, demonstrating a global awareness of the delivery network.

Figure A1 (d) presents time-conservative couriers. Here, the remaining delivery time (O1) is the primary factor, while the weights of the max distance from current customer to unfinished restaurants (R3) and the max distance from current restaurant to unfinished customers (R4) are comparable. This suggests that such couriers prioritize time sensitivity over spatial sensitivity and may be more willing to accept orders with flexible deadlines but longer distances.

Figure A1 (e) describes route-preferring couriers. The weight of the max distance from current customer to unfinished restaurants (R3) is notably high, while the max distance from current restaurant to unfinished customers (R4) and the max distance from current customer to unfinished customers (R5) exhibit a synergistic effect. These couriers focus on the spatial compatibility of new orders with their existing route, making route convenience a key factor in their decision-making. Therefore, when dispatching orders, the platform should customize rules for different categories of couriers based on their characteristics. For example, for couriers in Fig. A1b, the platform needs to control order dispatch frequency to fully leverage its efficiency advantages, thereby enhancing overall service quality. For groups that are more sensitive to geographic distribution (e.g. Fig. A1f), optimizing the range of order assignments can reduce long-distance delivery. By thoroughly understanding the characteristics and decision factors of different categories of couriers, the platform can enhance the accuracy and success rate of dispatching orders, consequently improving overall delivery efficiency. This can also increase courier satisfaction and reduce occurrences like couriers rejecting orders due to inappropriate dispatching.

C. Algorithmic Performance of LBB Method

In this section, the proposed LBB method is compared with the commercial solver Gurobi in solving the proposed

TABLE V
COMPARISON BETWEEN MILP MODEL AND LBBD METHOD

n	#Ins	CU	MILP				LBBD			
			Obj	Gap(%)	Time(s)	Opt	Obj	Gap(%)	Time(s)	Opt
10	A1	6	40.93	0.00	0.36		40.93	0.00	0.01	
	A2	3	44.06	0.00	0.78		44.06	0.00	0.29	
	A3	11	43.21	0.00	0.44		43.21	0.00	0.01	
	A4	6	36.82	0.00	0.48		36.82	0.00	0.01	
	A5	12	44.37	0.00	1.20		44.37	0.00	0.03	
Avg		8	41.88	0.00	0.65	5	41.88	0.00	0.07	5
20	B1	34	79.75	0.00	9.27		79.75	0.00	1.81	
	B2	22	84.75	0.00	13.65		84.75	0.00	5.54	
	B3	16	94.32	0.00	37.08		94.32	0.00	14.99	
	B4	26	89.74	0.00	34.96		89.74	0.00	15.28	
	B5	24	88.83	0.00	141.31		88.83	0.00	10.88	
Avg		24	87.48	0.00	47.25	5	87.48	0.00	9.70	5
30	C1	38	134.39	0.00	672.22		134.39	0.00	67.72	
	C2	24	142.54	0.00	2840		142.54	0.00	192.56	
	C3	53	114.64	0.00	90.35		114.64	0.00	15.80	
	C4	26	116.61	0.00	300.17		116.61	0.00	93.25	
	C5	53	123.30	0.13	3600		123.14	0.00	131.57	
Avg		39	126.30	0.03	1500.55	4	126.26	0.00	100.18	5
40	D1	39	186.78	3.96	3600		180.42	0.00	346.21	
	D2	49	165.34	1.23	3600		164.66	0.00	298.70	
	D3	33	179.86	0.00	1178.12		179.86	0.00	104.43	
	D4	65	179.77	0.70	3600		179.13	0.00	347.27	
	D5	27	257.75	0.05	3600		257.44	0.00	244.50	
Avg		43	193.90	1.20	3115.62	1	192.17	0.00	268.22	5

MILP. The computational results of the two models are shown in Table V. The first column (n) represents the number of orders, the second column (#Ins) represents different instances of the same scale, and the third column (CU) represents the total number of on-hand orders of the couriers. Subsequently, the optimal objective function value (Obj), the gap between the current solution and the optimal solution (Gap(%)), the computational time (Time(s)), and the number of instances achieving optimality for the same scale (Opt) are presented.

As shown in Table V, the performance of LBBD regarding the computational efficiency and solution quality is significantly better than those of Gurobi. In the 20 instances, the LBBD method solves all instances to optimality, while Gurobi can only solve 15 instances to optimality. When the number of orders is 30, the average gap of the MILP model solved by Gurobi is 0.03%. When the number of orders is 40, the average gap of the MILP model solved by Gurobi is 1.20%. In terms of computational time, the LBBD method is much faster than Gurobi. For all 20 instances, the average computation time of the MILP model by Gurobi is 1166.02 seconds, about 12 times that of the LBBD method (94.54 seconds). It can also be observed that as the number of orders increases, the computational time for solving the original MILP model is increased dramatically, while the LBBD method requires much less computational time to solve the same instances. For instances with 40 orders, the computational time achieved by the proposed LBBD method is 268 seconds, much less than that of Gurobi (i.e., 2115 seconds). In conclusion, LBBD outperforms Gurobi in both solution quality and computational efficiency.

Besides, further numerical experiments have been conducted to compare the performance of the proposed methodology and four heuristic rules. Due to the page limit, we put the comparison analysis into Appendix III of Supplementary Material.

TABLE VI
COMPARISON OF THE PROPOSED NO-GOOD CUTS AND ANALYTICAL CUTS

n	#Ins	No-good cuts				Analytical cuts			
		Obj	#Iter	Time(s)	#Cuts	Obj	#Iter	Time(s)	#Cuts
10	A1	40.93	4	0.00	30	40.93	4	0.00	60
	A2	44.06	70	1.10	486	44.06	24	0.40	304
	A3	43.21	3	0.10	19	43.21	3	0.10	38
	A4	36.82	3	0.00	13	36.82	3	0.00	26
	A5	44.37	6	0.10	29	44.37	4	0.10	38
Avg		41.88	17	0.26	115	41.88	8	0.12	93
20	B1	79.75	53	1.50	457	79.75	31	0.90	540
	B2	84.75	62	17.00	589	84.75	75	10.00	1336
	B3	94.32	238	6.60	2534	94.32	210	6.40	4652
	B4	89.74	236	11.10	2411	89.74	79	4.20	1584
	B5	88.83	242	21.40	2420	88.83	281	25.80	5624
Avg		87.48	166	11.52	1682	87.48	135	9.46	2747
30	C1	134.39	473	35.90	7384	134.39	210	9.70	6710
	C2	142.54	760	189.30	12952	142.54	930	110.60	32218
	C3	114.64	69	2.70	1185	114.64	57	1.70	1904
	C4	116.61	559	75.20	9054	116.61	483	75.10	16030
	C5	123.14	3281	1130.20	48513	123.14	956	133.10	28780
Avg		126.26	1028	286.66	15818	126.26	527	66.04	17128
40	D1	180.42	1192	340.66	24413	180.42	1772	339.01	73766
	D2	164.66	7028	5777.20	150722	164.66	833	298.70	34904
	D3	179.86	918	78.80	21025	179.86	1400	104.43	63410
	D4	179.13	736	230.10	13941	179.13	860	347.27	32192
	D5	257.44	9009	954.90	197365	257.44	1340	244.50	88026
Avg		192.30	3777	1476.33	81493	192.30	1241	266.78	58460

D. Algorithmic Performance of the Benders Cuts

Preliminary experiments have demonstrated that the no-good cuts have limited effects on the convergence of the algorithm and solution quality. These cuts reduce the solution efficiency and make it harder to decrease the search space. Therefore, in order to increase the constraints of the cuts on our problem to improve the algorithmic performance, we further propose the new analytical Benders cuts. Based on the structural analysis of the problem, the analytical Benders cuts are designed to more effectively capture the key constraints, thereby shrinking the feasible solution space and accelerating convergence in each iteration. To evaluate the performance of the newly proposed analytical Benders cuts, we utilize 20 instances with orders of 10, 20, 30, 40 for experiments. We use both the no-good cuts and the analytical Benders cuts, and compare the computational results of the two kinds of cuts. The results are shown in Table VI. Additionally, several indicators are detailed in the table, including the number of iterations (#Iter) and the number of Benders cuts added to the master problem (#Cuts). From this table, it can be observed that both cuts perform well in the LBBD method proposed in this section. However, the performance of the analytical Benders cuts is still superior to that of the no-good cuts, in terms of computational time. The average computation time for the analytical Benders cut is 85.60s, which is about 81% faster than the no-good cut (443.69s). The average number of iterations and cuts for the analytical Benders cut are 477.75 and 19607.10, respectively (both lower than the no-good cut). The computational results verify that the analytical Benders cut is more efficient than the no-good cut.

E. Sensitivity Analysis

In this section, we further investigate the impacts of different cost parameters in the objective function on the optimal dispatch solutions through sensitivity analysis. We conduct the sensitivity analysis over problem scales of $n = 10, 20, 30$ and 40. The total cost of the platform is composed of

TABLE VII

THE IMPACT OF THE COST PER ORDER OF DEDICATED COURIERS

n	α	TC	DC	CC	RC	OC	ND	NG
10	4	37.05	28	8.65	0.40	0	7	3
	4.5	40.55	31.50	8.65	0.40	0	7	3
	5	44.05	35	8.65	0.40	0	7	3
	5.5	47.55	38.50	8.65	0.40	0	7	3
	6	51.05	42.00	8.65	0.40	0	7	3
20	4	73.10	48	19.58	2.62	2.88	12	8
	4.5	79.10	54	19.58	2.62	2.88	12	8
	5	84.75	55	24.19	2.67	2.88	11	9
	5.5	90.25	60.50	24.19	2.67	2.88	11	9
	6	95.75	66	24.19	2.67	2.88	11	9
30	4	114.37	84	22.14	1.61	6.62	21	9
	4.5	124.39	90	26.01	1.69	6.69	20	10
	5	134.39	100	26.01	1.69	6.69	20	10
	5.5	144.06	104.50	31.14	1.74	6.69	19	11
	6	153.56	114	31.14	1.74	6.69	19	11
40	4	139.47	112	22.12	3.72	1.63	28	12
	4.5	152.73	117	30.34	3.76	1.63	26	14
	5	165.25	120	39.67	3.95	1.63	24	16
	5.5	175.46	121	48.82	4.01	1.63	22	18
	6	188.01	126	56.28	4.10	1.63	21	19

four parts, namely, the payment of dedicated couriers, the payment of crowdsourced couriers, the cost due to order rejection by the crowdsourced couriers, and the penalty cost due to overtime delivery. All the instances are solved by the LBBD proposed in this paper. The table shows the average value under each scenario. TC, DC, CC, RC, OC refer to the total cost, the payment of dedicated couriers, the payment of crowdsourced couriers, the operational cost due to order rejections by crowdsourced couriers and the penalty cost due to overtime delivery. ND and NG indicate the number of orders assigned to dedicated couriers and crowdsourced couriers respectively.

1) *Fees Paid to Dedicated Couriers for Each Order Delivery*: We assume that the delivery payment for dedicated couriers varies within the range $\{4, 4.5, 5, 5.5, 6\}$, while other parameters remain consistent with those outlined in part A of Section V. Subsequently, we record the cost values and analyze the changes in the objective function values, with results presented in Table VII. Note that when α is changing, the proposed LBBD method obtain optimal solutions for all instances. First, as the delivery payment for dedicated couriers gradually increases, there is a noticeable decline in the orders allocated to dedicated couriers, contrasted by an increase in orders assigned to crowdsourced couriers. However, it is observed that the penalty costs for late delivery is also increased a bit, which is caused by the increase of orders allocated to crowdsourced couriers. This is because there are orders on hand by the crowdsourced couriers that may incur overtime delivery for new assigned orders. It indicates that, although it may reduce the service quality by transforming some orders from the dedicated couriers to the crowdsourced couriers, it is a trade-off solution among the multiple performance criteria. If the total number of dedicated couriers becomes smaller and the payment have to be added, it is much wiser for platform to hire more crowdsourced couriers while considering their preferences at the same time.

2) *Fees Paid to Crowdsourced Couriers per Unit Kilometer*: Assuming that the payment per unit kilometer of crowdsourced couriers varies within $\{1, 1.25, 1.5, 1.75, 2\}$, the settings of other parameters are the same as those in part A of Section V,

TABLE VIII

THE IMPACT OF THE PAYMENT PER UNIT KILOMETER OF CROWDSOURCED COURIERS

n	β	TC	DC	CC	RC	OC	ND	NG
10	1	41.17	35	5.76	0.40	0	7	3
	1.25	42.61	35	7.21	0.40	0	7	3
	1.5	44.05	35	8.65	0.40	0	7	3
	1.75	45.49	35	10.09	0.40	0	7	3
	2	46.94	35	11.53	0.40	0	7	3
20	1	76.68	55	16.12	2.67	2.88	11	9
	1.25	80.72	55	20.16	2.67	2.88	11	9
	1.5	84.75	55	24.19	2.67	2.88	11	9
	1.75	88.36	60	22.85	2.62	2.88	12	8
	2	91.63	60	26.11	2.62	2.88	12	8
30	1	123.47	85	29.98	1.80	6.69	17	13
	1.25	129.38	95	25.95	1.74	6.69	19	11
	1.5	134.39	100	26.01	1.69	6.69	20	10
	1.75	138.73	100	30.35	1.69	6.69	20	10
	2	142.75	105	29.52	1.61	6.62	21	9
40	1	148.05	105	37.52	3.89	1.63	21	19
	1.25	157.43	105	46.90	3.89	1.63	21	19
	1.5	165.07	120	39.67	3.77	1.63	24	16
	1.75	171.15	130	35.40	4.12	1.63	26	14
	2	174.42	145	24.33	3.46	1.63	29	11

TABLE IX

THE IMPACT OF THE COST COEFFICIENT OF REJECTION PROBABILITY

n	c_p	TC	DC	CC	RC	OC	ND	NG	Overtime(min)
10	3	63.24	25	24.42	2.87	10.95	5	5	9.22
	4	64.20	25	24.42	3.83	10.95	5	5	9.22
	5	65.09	30	22.91	1.87	10.32	6	4	11.83
	6	65.46	30	22.91	2.24	10.31	6	4	11.83
	3	93.55	40	26.51	24.16	2.88	8	12	0.00
20	4	100.59	45	26.86	25.85	2.88	9	11	0.00
	5	106.50	50	27.70	24.02	4.77	10	10	2.03
	6	111.30	50	27.70	28.83	4.77	10	10	2.03
	3	134.27	65	52.64	13.93	2.70	13	17	4.48
	4	139.72	65	52.73	19.06	2.93	13	17	4.48
30	5	144.60	80	44.85	15.35	4.40	16	14	4.48
	6	147.05	85	42.87	15.82	3.36	17	13	5.13
	3	187.05	105	50.11	24	7.94	21	19	7.82
	4	195.73	110	47.45	30.34	7.94	22	18	7.82
	5	202.76	115	53.16	26.78	7.82	23	17	7.82
40	6	208.06	115	57.26	27.98	7.82	23	17	7.82

and the results are shown in Table VIII. From the table, it can be seen that as the payment per unit kilometer of crowdsourced couriers gradually increases, the orders allocated to crowdsourced couriers decrease and the orders allocated to dedicated couriers increase. This indicates that the system tends to prioritize dedicated couriers with relatively low cost to control the overall cost. When $\beta = 2$, the dedicated couriers take at least 60% of the orders. And the overtime decreases, reflecting the improvement of the dedicated couriers' delivery efficiency. Moreover, with the increase of β , fewer and fewer orders are allocated to crowdsourced couriers, and the cost due to the order rejection by crowdsourced couriers also decreases.

3) *Cost Coefficient of the Rejection Probability*: Assuming the cost coefficient of the rejection probability varies within the range $\{3, 4, 5, 6\}$, the upper limit for the latest delivery time of orders is set at 1800 seconds. Here, we relax the original assumption and let the customer set served by the crowdsourced couriers identical to that of the dedicated couriers. The other parameters remain consistent with those outlined in part A of Section V. Subsequently, various cost values are recorded, along with the overtime duration for the dedicated couriers, as presented in Table IX. It is evident that the number of orders allocated to dedicated couriers is gradually increasing, while those assigned to crowdsourced

TABLE X

THE IMPACT OF THE PENALTY COST COEFFICIENT OF UNIT OVERTIME

n	cd	TC	DC	CC	RC	OC	ND	NG	Overtime (min)
10	1	49.01	35.0	12.27	1.50	0.24	7	3	0.24
	2	51.25	30.0	18.75	2.02	0.48	6	4	0.24
	3	51.50	30.0	18.75	2.02	0.72	6	4	0.24
	4	51.74	30.0	18.75	2.02	0.96	6	4	0.24
20	1	94.08	45	40.41	8.27	0.40	9	11	0.4
	2	94.26	45	40.41	8.27	0.58	9	11	0.29
	3	94.55	45	40.41	8.27	0.87	9	11	0.29
	4	94.68	40	44.87	8.65	1.16	8	12	0.29
30	1	154.95	80	37.60	25.87	11.48	16	14	11.48
	2	166.15	75	38.77	30.31	22.06	15	15	11.03
	3	177.18	75	38.77	30.31	33.10	15	15	11.03
	4	188.22	75	38.77	30.31	44.14	15	15	11.03
40	1	203.18	75	86.29	29.32	12.57	15	25	12.57
	2	213.56	70	91.19	30.94	21.43	14	26	10.72
	3	229.84	70	91.19	30.94	37.71	14	26	10.72
	4	242.41	70	91.19	30.94	50.28	14	26	10.72

couriers are decreasing. This indicates that when the platform is reluctant to allocate orders to crowdsourced couriers to avoid low delivery efficiency due to a high order rejection rate, i.e., a high cost coefficient for the rejection rate, the platform will force the dedicated couriers to deliver these orders, resulting in an imbalance in order dispatch. Furthermore, the overtime for orders delivered by dedicated couriers tends to increase, which may reduce customer satisfaction significantly. It is indicated that, for the platform, it is wiser to hire more crowdsourced couriers to improve delivery efficiency. However, if the platform becomes too “risk averse” to the order rejection rate by crowdsourced couriers, it may inhibit the order assignment to the crowdsourced couriers and reduce the delivery system’s efficiency.

4) *Penalty Cost Coefficient of Unit Overtime*: We set the penalty cost coefficient for late delivery varies within the range {1,2,3,4}, the upper limit for the latest delivery time of orders is set as 1800 seconds. Here, we fix the cost coefficient of the order rejection rate as 5. We again relax the original assumption and let the customer set that can be served by the crowdsourced couriers is identical to that of the dedicated couriers. Other parameters remain consistent with those outlined in part A of Section V. The values of the various costs and the number of orders assigned to different types of couriers are presented in Table X. As the tardiness penalty cost coefficient c_d increases from 1 to 4, the total cost TC also increases progressively, while the total overtime decreases. The number of orders allocated to dedicated couriers shows a downward trend, whereas the cost associated with crowdsourced couriers shows an upward trend. This indicates that even the platform sets a high cost coefficient regarding order rejection rate to avoid re-dispatching, it is still wiser to allocate more orders to crowdsourced couriers to guarantee on-time delivery of orders and minimize total delivery delays, by considering the couriers’ preference and controlling the potential risk that the crowdsourced couriers choose to reject servicing the assigned orders. In addition, it is found that the cost of potential order rejection does not increase too much. Therefore, it is identified that the proposed model helps the platform to realize trade-off between the total cost, service quality and the crowdsourced couriers’ welfare.

TABLE XI

COMPARISON WITH AND WITHOUT THE CONSIDERATION OF COURIERS’ ORDER ACCEPTANCE PREFERENCE

n	#Ins	DC_0	CC_0	OC_0	Z_0	RC_0	DC_1	CC_1	OC_1	Z_1	RC_1
10	A1	25.00	14.02	3.69	42.72	9.53	25.00	14.02	3.69	42.71	10.38
	A2	30.00	18.36	1.03	49.39	2.65	30.00	18.36	0.84	49.20	4.78
	A3	35.00	7.31	0.00	42.31	5.42	35.00	7.31	0.00	42.31	7.80
	A4	25.00	11.02	3.14	39.16	4.78	25.00	11.02	3.14	39.16	7.76
	A5	35.00	8.55	0.00	43.55	5.43	35.00	8.55	0.00	43.55	5.68
20	B1	55.00	22.77	0.00	77.77	12.81	55.00	22.77	0.00	77.77	16.01
	B2	65.00	18.72	3.38	87.10	11.08	55.00	24.19	2.89	82.08	18.41
	B3	60.00	36.65	1.06	97.70	2.01	60.00	36.65	1.06	97.71	8.85
	B4	75.00	15.90	0.00	90.91	3.12	70.00	18.80	0.00	88.80	9.91
	B5	60.00	33.10	6.31	99.42	4.16	60.00	33.10	6.31	99.42	9.34
30	C1	100.00	26.01	7.42	133.43	10.29	100.00	26.01	7.20	133.21	17.03
	C2	100.00	40.00	5.90	145.80	2.75	100.00	39.90	5.90	145.80	9.89
	C3	85.00	28.35	1.16	114.52	16.50	80.00	31.51	0.30	111.81	26.91
	C4	85.00	31.13	1.49	117.62	13.17	85.00	31.13	1.22	117.35	16.85
	C5	85.00	36.54	0.00	121.54	10.93	85.00	36.54	0.00	121.54	20.21
40	D1	120.00	62.18	6.81	188.99	12.9	120.00	62.18	6.81	188.99	22.04
	D2	120.00	39.68	0.98	160.66	11.88	120.00	39.68	0.98	160.66	29.90
	D3	120.00	70.26	8.41	198.67	8.46	120.00	70.26	8.41	198.67	16.43
	D4	100.00	99.27	19.58	218.85	13.83	100.00	99.27	19.58	218.85	26.53
	D5	120.00	48.28	107.04	275.32	11.09	120.00	48.28	107.04	275.32	16.13
Avg		75.00	33.41	8.87	117.27	8.63	74.00	33.98	8.87	116.85	15.04

Besides the aforementioned sensitivity analysis, to evaluate the impact of the prediction error on the optimal solution, we also conduct a sensitivity analysis to observe the variance in the objective values and allocation results. Due to the page limit, we put this part into Appendix V of Supplementary Material.

F. Importance of Considering Couriers’ Order Preferences

In this section, we analyze the importance of considering couriers’ order preferences during the order dispatching decision making process. For the optimal solution obtained by the proposed model, the sum of the payment for dedicated couriers DC_0 , crowdsourced couriers CC_0 and the overtime penalty OC_0 is represented by Z_0 , and the cost of potential rejected orders by the crowdsourced couriers is represented by RC_0 . At the same time, we consider the benchmark where couriers’ order preference is not considered, and the sum of the three costs, i.e. the payments of dedicated couriers DC_1 , crowdsourced couriers CC_1 , and overtime cost OC_1 , is expressed in RC_1 , and the cost of rejected orders under the benchmark is expressed as RC_1 . For an in-depth analysis, we select 15 cases with the number of orders $n = \{10, 20, 30, 40\}$. In these instances, the upper limit of latest delivery time is set to 1800s, and the cost coefficient of the potential number of rejected orders is set into a moderate level, i.e., $\beta = 3$. Other parameters are the same as those in part A of Section V.

The results are presented in Table XI, which illustrates the cost differences between the proposed model and the benchmark. We can find that the cost due to potential rejected orders under the benchmark is nearly twice of that of the proposed model (i.e., considering the couriers’ order preference), while the total operational cost increased under the proposed model is lower than 1%. This indicates that incorporating the individual preferences of crowdsourced couriers into the model does not lead to notable operational cost increase, instead, it significantly increases the dispatching reliability by improving the couriers’ willingness to accepting the orders

assigned to them. Furthermore, the model that accounts for courier preferences not only reduces the total cost but also enhances couriers' welfare, which is crucial for improving delivery efficiency and service quality. Although in certain specific cases, the impact of couriers' preferences on costs may not be obvious. But in general, the model based on couriers' order preferences demonstrates a clear advantage in optimizing couriers dispatch, reducing operational costs, and enhancing couriers' welfare. Therefore, for on-demand delivery platforms, integrating courier preference information into the courier dispatch system is a vital strategy for achieving more efficient and cost-effective operations, as well as an important pathway to societal sustainable operations.

VI. CONCLUSION

To solve the problem of inadequate consideration of courier behavior characteristics in traditional on-demand delivery courier dispatch system, our research proposes a novel data-driven method framework that incorporates predictive analysis into the mathematical optimization to address the uncertainty of crowdsourced couriers' willingness to accept orders in the on-demand delivery dispatch system. Based on the large-scale order data from the Meituan platform, this study identifies and extracts eight features that influence couriers' order acceptance behavior, and applies an XGBoost model for order acceptance probability prediction. Afterwards, a mixed integer programming model is formulated to optimize the order dispatching decisions by incorporating the predicted probability of accepting specific orders to the specific couriers, which is further solved by the LBB algorithm. An efficient analytical cut is built to accelerate the computational time for optimality. Numerical experiments show the substantial benefits of this proposed model in enhancing dispatch system's efficiency, reducing rejection rates, and increasing the crowdsourced couriers' welfare. In addition to providing a novel order dispatching strategy for on-demand delivery platforms, this study offers new insights on courier behavior characteristics.

This study has limitations, and can be extended for future work. First, to increase the accuracy and flexibility of courier dispatching strategies, it is essential to consider the integration of real-time information (e.g. weather, traffic situation) to construct an adaptive dynamic predictive model that can continuously update courier preferences. To be specific, the posterior probability when more than one orders dispatched at a time can be further analyzed in the future. Second, to model the delivery time more accurately, the assumptions could be further relaxed. For example, the varying speeds of individual couriers and more accurate calculation of the distance between two nodes (rather than Manhattan distance) based on the information of map service provides (e.g. AutoNavi API) will be considered. Third, the exploration of more advanced artificial intelligence algorithms, such as reinforcement learning, deep reinforcement learning, or optimization algorithms based on graph neural networks, could be beneficial. These methodologies have the potential to significantly improve computational efficiency and dispatch effectiveness in complex

scenarios through continuous learning and large-scale data training.

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Xuting Sun received the Ph.D. degree in industrial and systems engineering from The Hong Kong Polytechnic University, Hong Kong, in 2018. She is currently an Associate Professor with the SILC Business School, Shanghai University, Shanghai, China. She has publications in top-tier SCI/SSCI journals, such as *Transportation Research—Part B: Methodological*, *Transportation Research—Part C: Emerging Technologies*, *Transportation Research—Part E: Logistics and Transportation Review*, *Decision Sciences*, *Decision Support Systems*, *IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT*, *IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS*, *Reliability Engineering and System Safety*, *Computers and Industrial Engineering*, and *Journal of Air Transport Management*. Her research interests include smart and sustainable logistics operations, big data analytics, and risk management. She is a member of the Advisory Board of *Transportation Research—Part E: Logistics and Transportation Review* and *IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT*. She was a Guest Editor of *IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT* and *Journal of Air Transport Management*.



Yue Hu is currently pursuing the M.S. degree in management science and engineering with Shanghai University. She has published her work in top-tier journals, such as *Transportation Research—Part E: Logistics and Transportation Review* and *Reliability Engineering and System Safety*. Her research interests include intelligent optimization and scheduling, with a particular focus on applications in the field of instant delivery.



Jiage Huo received the Ph.D. degree in industrial and systems engineering from The Hong Kong Polytechnic University, Hong Kong, China, in 2020. She is currently a Lecturer with the SILC Business School, Shanghai University, Shanghai, China. She has publications in top-tier journals, such as *IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT*, *IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS*, *International Journal of Production Research*, and *Journal of Business Research*. Her research interests include production scheduling, optimization methods, and green innovation.